

Groups

Modular Arithmetic :-

The wrapping around of a number after reaching certain required value is called modulus.

$$a \equiv b \pmod{m}$$

It is a central mathematical concept used in cryptography. Here, congruence ($\equiv$) is used instead of equality ($=$).

eg. $16 \equiv 4 \pmod{12}$

$$13 \equiv 1 \pmod{12}$$

$$23 \equiv 3 \pmod{10}$$

$$\begin{array}{r} 10 \overline{) 23} \\ \underline{20} \\ 3 \end{array} \rightarrow \text{Remainder.}$$

$$\Rightarrow a = km + b$$

eg. $23 = 10(2) + 3$

examples

Q. 1.

Solve $(23)^3 \pmod{30}$

$$23 \equiv -7 \pmod{30}$$

$$\therefore (23)^3 \pmod{30} \equiv (-7)^3 \pmod{30}$$

$$\equiv (-7)^2(-7) \pmod{30}$$

$$\equiv 49(-7) \pmod{30}$$

$$\equiv -343 \pmod{30}$$

$$\equiv -13 \pmod{30}$$

$$\therefore (23)^3 \pmod{30} \equiv -13$$

Q.2. Find $(31)^{500} \pmod{30}$

$$\rightarrow 31 \equiv 1 \pmod{30}$$

Thus,

$$\begin{aligned} (31)^{500} \pmod{30} &= (1)^{500} \pmod{30} \\ &= 1 \pmod{30} \end{aligned}$$

Thus,

$$(31)^{500} \pmod{30} = 1$$

Q.3 Find the remainder of $(242)^{329} \pmod{243}$.

$$\rightarrow 242 \equiv (-1) \pmod{243}$$

Thus,

$$(-1)^{329} \pmod{243} = (242)^{329} \pmod{243}$$

$$(-1) \pmod{243} = (242)^{329} \pmod{243} \equiv -1$$

Thus,

$$(242)^{329} \pmod{243} \equiv -1$$

$\therefore$ The remainder is, 242

Q.4. $(11)^7 \pmod{13}$ find

$\rightarrow$

$$11 \equiv -2 \pmod{13}$$

$$(11)^7 \pmod{13} = (-2)^7 \pmod{13}$$

$$= (-128) \pmod{13}$$

$$= (-11) \pmod{13}$$

Thus,

$$(11)^7 \pmod{13} \equiv -11$$

* Prime Number Theory (Euclid Theorem) :-

A positive integer greater than 1 is said to be prime if it is divisible by 1 or itself (only two factors). The numbers other than prime are called composite numbers.

* Fundamental Theorem of Arithmetic :-

Every positive integer greater than 1 can be written uniquely as a prime or as a product of two or more prime numbers.

eg.

$$13 = 13 \times 1$$

$$100 = 2 \times 2 \times 5 \times 5$$

* Congruence :-

Two integers a & b are congruent module m if and only if they have same remainder when divided by m .

$$a \equiv b \pmod{m}$$

Note:- ① $a \equiv b \pmod{m}$ means $a \pmod{m} = b \pmod{m}$

② $a \equiv b \pmod{m}$ iff m divides $a-b$.

③ Congruence mod m is an equivalence relation.

* Residue Classes :-

Let ' r ' be an integer then the residue class of ' r ' under mod m is $[r] = \{x/x \in \mathbb{Z} \text{ and } x \equiv r \pmod{m}\}$

e.g. ① Residue class modulo 4

$$[r] = \{x/x \in \mathbb{Z} \text{ and } x \equiv r \pmod{4}\}$$

$$② [0] = \{0, 4, 8, 12, 16, \dots\}$$

$$[1] = \{1, 5, 9, 13, 17, \dots\}$$

$$[2] = \{2, 6, 10, 14, \dots\}$$

$$[3] = \{3, 7, 11, 15, 19, \dots\}$$

$\therefore$ Residue class of modulo 4 is,

$$\therefore \mathbb{Z}_4 = \{[0], [1], [2], [3], \dots, 4 \equiv 0 \pmod{4}\}$$

Thus,

$$\mathbb{Z}_m = \{[0], [1], [2], [3], \dots, [m-1]\}$$

* Addition of Residue Classes :-

Let $[a], [b] \in \mathbb{Z}_m$ then

$$[a] +_m [b] = [r]$$

where,

r is remainder when ' $a+b$ ' is divided by m .

e.g. ① $\mathbb{Z}_4 = \{[0], [1], [2], [3]\}$

$$[1] +_4 [2] = [3]$$

$$[2] +_4 [3] = [1]$$

Multiplication of Residue Class :-

Let.

$$[a], [b] \in \mathbb{Z}_m \text{ then } [a] \times_m [b] = [r]$$

where,

r is remainder when ' $a \times b$ ' is divided by m .

eg. ① $\mathbb{Z}_5 = \{[0], [1], [2], [3], [4]\}$

$$[2] \times_5 [3] = [1]$$

$$[3] \times_5 [4] = [2]$$

Fermat's Theorem :-

If p is an prime integer and a is any integer then

$$a^p \equiv a \pmod{p}$$

(or)

$$a^{p-1} \equiv 1 \pmod{p}$$

(or)

$$\text{g.c.d}(a, p) = 1$$

eg. ① Using Fermat's theorem find $(3)^{81} \pmod{7}$

→ Here,

$$a = 3, p = 7$$

using Fermat's theorem

$$a^{p-1} \equiv 1 \pmod{p}$$

$$\therefore (3)^{7-1} \equiv 1 \pmod{7}$$

$$(3)^6 \equiv 1 \pmod{7}$$

Now,

$$\begin{aligned}
 (3)^{31} \bmod 7 &\equiv (3)^{(7 \times 4 + 3)} \bmod 7 \\
 &\equiv (3^7)^4 \cdot (3)^3 \bmod 7 \\
 &\equiv (1)^4 \cdot 3 \bmod 7 \\
 &\equiv 3 \bmod 7
 \end{aligned}$$

Thus,

$$(3)^{31} \bmod 7 = 3$$

Q.2 Find the remainder when $(21)^{115}$ is divided by 29 using Fermat's Theorem. (or) Solve $(21)^{115} \bmod (29)$.

By Fermat's Theorem,
 $a^{p-1} \equiv 1 \pmod{p}$

Here,

$$a = 21, \quad p = 29$$

$$(21)^{28} \equiv 1 \pmod{29}$$

$$(21)^{28} \equiv 1 \pmod{29}$$

Now,

$$\begin{aligned}
 (21)^{115} \bmod (29) &= (21)^{28 \times 4 + 3} \bmod (29) \\
 &= (21)^{28 \times 4} \cdot (21)^3 \bmod (29) \\
 &= ((21)^{28})^4 \cdot (21)^3 \bmod (29) \\
 &= 1 \cdot (9261) \bmod (29) \\
 &= 10 \bmod (29)
 \end{aligned}$$

29	9261

Thus, if $(21)^{115}$ is divided by 29 remainder is 10.

Q. ③ What is the remainder when $(7)^{2001}$ is divided 5.

By Fermat's Theorem,
 $a^{p-1} \equiv 1 \pmod{p}$

Here,

$$a = 7, \quad p = 5$$

$$(7)^{5-1} \equiv 1 \pmod{5}$$

$$(7)^4 \equiv 1 \pmod{5}$$

Now,

If $a \equiv b \pmod{n}$, $c \equiv d \pmod{n}$ then.

$$a \cdot c \equiv b \cdot d \pmod{n}$$

$$\therefore 7^4 \cdot 7^4 \equiv 1 \cdot 1 \pmod{5}$$

$$7^8 \equiv 1 \pmod{5}$$

$$7^8 \cdot 7^4 \equiv 1 \pmod{5}$$

$$7^{12} \equiv 1 \pmod{5}$$

⋮

$$7^{2001} \pmod{5} \equiv (1)^{2001} \cdot (7)^1 \pmod{5}$$

$$7^{2001} \pmod{5} \equiv 7 \pmod{5}$$

$$\equiv 2 \pmod{5}$$

Thus, the remainder is 2.

11.11

Q. ④ $(2)^{50} \pmod{17}$

By Fermat's Theorem.

Q. ⑤ $(4)^{532} \pmod{11}$

Q. ⑥ $(2)^{16} \pmod{17}$

Q. ⑦ $(9)^{799} \pmod{73}$

Q. ⑧ Find the remainder when 2^{100} is divided by 15.

H.W

Q.4.

$$(2)^{50} \text{ mod } 17$$

→

By Fermat's Theorem,

Here, $n=17$, $a=2$

$$a^{n-1} \equiv 1 \text{ mod } n$$

$$2^{16} \equiv 1 \text{ mod } 17$$

$$2^{16} \equiv 1 \text{ mod } 17$$

$$2^{50} = 2^{16 \times 3 + 2} \rightarrow (2^{16})^3 \cdot 2^2$$

$$\therefore 2^{50} \text{ mod } 17 \equiv (2^{16})^3 \cdot 2^2 \text{ mod } 17$$

$$\equiv (2^{16})^3 \text{ mod } 17 \times 2^2 \text{ mod } 17$$

$$\equiv 1 \cdot 2^2 \text{ mod } 17$$

$$\therefore 2^{50} \text{ mod } 17 \equiv \underline{\underline{4}}$$

Thus the remainder is 4

H.W

Q.5.

$$4^{532} \text{ mod } 11$$

→

By Fermat's Theorem,

$$a \neq 4, p = 11$$

$$\therefore 4^{10} \equiv 1 \text{ mod } 11$$

$$4^{10} \equiv 1 \text{ mod } 11$$

$$4^{532} \equiv (4^{10})^{53} \cdot 4^2 \text{ mod } 11$$

$$\equiv (1 \cdot \text{mod } 11, 4^2 \text{ mod } 11) \text{ mod } 11$$

$$\equiv (1 \times 16 \text{ mod } 11) \text{ mod } 11$$

$$\equiv 16 \text{ mod } 11$$

$$\equiv 5 \text{ mod } 11$$

$$4^{532} \text{ mod } 11 \equiv 5$$

$\therefore$ Thus the remainder is 5

Q.6

$$e^{16} \text{ mod } 17$$

Here, $a = e$, $p = 17$

By Fermat's Theorem,

$$a^{p-1} \equiv 1 \text{ mod } p$$

$$e^{17-1} \equiv 1 \text{ mod } 17$$

$$e^{16} \equiv 1 \text{ mod } 17$$

$$e^{16} \text{ mod } 17 \equiv (1 \text{ mod } 17) \text{ mod } 17$$

$$\therefore e^{16} \text{ mod } 17 \equiv 1$$

$\therefore$ Remainder is 1.

Q.7

Q.7

$$(g)^{794} \text{ mod } 73$$

Here, $a = g$, $p = 73$.

By Fermat's theorem,

$$a^{p-1} \equiv 1 \text{ mod } p$$

$$g^{73-1} \equiv 1 \text{ mod } 73$$

$$g^{72} \equiv 1 \text{ mod } 73$$

$$\therefore g^{794} = g^{72 \times 11 + 2} \rightarrow (g^{72})^{11} \cdot g^2$$

$$\therefore g^{794} \equiv (g^{72})^{11} \cdot g^2 \text{ mod } 73$$

$$\equiv (g^{72})^{11} \text{ mod } 73 \times g^2 \text{ mod } 73$$

$$\equiv 1 \cdot 81 \text{ mod } 73$$

$$\equiv 81 \text{ mod } 73$$

$$\equiv +8 \text{ mod } 73$$

$$\therefore g^{794} \equiv 8 \text{ mod } 73$$

$$\therefore g^{794} \text{ mod } 73 \equiv 8 \text{ mod } 73$$

$\therefore$ Thus, the remainder is 8.

Q.3. Find the remainder when 2^{100} is divided by 15.
 $\rightarrow 2^{100} \bmod 15.$

Here, $p=15$, $a=2$.

$\therefore$ By Fermat's Theorem,

$$a^{p-1} \equiv 1 \bmod p.$$

$$2^{15-1} \equiv 1 \bmod 15$$

$$2^{14} \equiv 1 \bmod 15$$

$$\therefore 2^{100} = 2^{14 \times 7 + 2} = (2^{14})^7 \cdot 2^2$$

$$\therefore 2^{100} \equiv (2^{14})^7 \cdot 2^2 \bmod 15$$

$$\equiv 1 \cdot 2^2 \bmod 15$$

$$\equiv 4 \bmod 15.$$

$$\therefore 2^{100} \equiv 4 \bmod 15$$

$$\therefore 2^{100} \bmod 15 \equiv 4.$$

Thus, the remainder is 4.

* Algebraic Structure / Binary Algebra :-

A non-empty set G equipped with one binary operation $*$ is called binary algebra. (or) algebraic structure.

i.e., it must be closed under that operation.

It is denoted by $(G, *)$

eg. $(\mathbb{N}, +)$, $(\mathbb{Z}, -)$, $(\mathbb{Q}, \times)$

* Semi-Group :-

An algebraic structure $(G, *)$ is called semigroup if it follows associative property.

i.e., $(a * b) * c = a * (b * c)$

eg. $(\mathbb{N}, +)$, $(\mathbb{Z}, \times)$

* Monoid :-

The semi-group is called monoid if there exist an identity element 'e' in G such that $e * a = a * e = a$, for all $a \in G$.

eg. 1) $(\mathbb{N}, \times)$. It is monoid.
2) $(\mathbb{N}, +)$. It is not monoid.

H.W.
Q.

Determine whether the set with binary operation is a semi-group (or) monoid. : $(\mathbb{Z}^+, *)$ such that $a * b = \max(a, b)$ for all $a, b \in \mathbb{Z}^+$

→

The algebraic structure is $(\mathbb{Z}^+, *)$ where

$$a * b = \max(a, b)$$

i) Closure:-

for any $a, b \in \mathbb{Z}^+$, $a * b \in \mathbb{Z}^+$
 $\therefore \mathbb{Z}^+$ is closed under $*$

ii) Associative:-

for any three elements a, b, c ,
 $a * (b * c) = (a * b) * c$

where $*$ results maximum of a, b & c which are elements of $\mathbb{Z}^+$ thus, $\mathbb{Z}^+$ follows associative

iii) Identity:-

for all $a \in G$

H.W.
Q.

Show that set N is monoid with respect to multiplication.

* Group :-

A monoid $(G, *)$ with identity element is called a group if to each element $a \in G$ there exist an element $b \in G$ such that,

$$a * b = b * a = e$$

where b is called inverse of a , denoted by a^{-1}

(or)

An algebraic structure is called groups when it follows following properties,

- 1) Closure
- 2) Associative
- 3) Existence of identity
- 4) Inverse

eg. $(\mathbb{Z}, +)$, $(\mathbb{R}, +)$, $(\mathbb{Q}, \times)$

Q.1. Prove that $(\mathbb{Z}_4, +_4)$ is group.

→ Here,

$\mathbb{Z}_4$ denotes equivalence class generated by equivalence relation congruent modulo 4

$$\begin{aligned} \therefore \mathbb{Z}_4 &= \{[0], [1], [2], [3]\} \\ &= \{0, 1, 2, 3\} \end{aligned}$$

The cayley's composition table under binary operation modulo 4 of addition is,

$+_4$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

from above table,

1) Closure :-

for all $a, b \in \mathbb{Z}_4$

$a +_4 b \in \mathbb{Z}_4$

$\therefore \mathbb{Z}_4$ is closed under $+_4$.

2) Associative :-

for $a, b, c \in \mathbb{Z}_4$

$$a +_4 (b +_4 c) = (a +_4 b) +_4 c$$

$\therefore \mathbb{Z}_4$ is associative.

3) Identity :-

Here, 0 is identity

$$\text{i.e., } 0 +_4 0 = 0$$

$$1 +_4 0 = 1$$

$$2 +_4 0 = 2$$

$$3 +_4 0 = 3$$

4) Existence of inverse :-

$$0 +_4 0 = 0$$

$$1 +_4 3 = 0$$

$$2 +_4 2 = 0$$

$$3 +_4 1 = 0$$

$\therefore$ Inverse exist for every element of $\mathbb{Z}_4$

$\therefore (\mathbb{Z}_4, +_4)$ is a group.

$\therefore (Z_4, +_4)$ is a group.

Q.2 Prove that $(Z_4, \times_4)$ is not a group.

Note 1) Every group has a unique identity. ~~every~~ group
 2) Every element in a group has a unique inverse.

→ Here,

$$Z_4 = \{[0], [1], [2], [3]\}$$

$$\therefore Z_4 = \{0, 1, 2, 3\}$$

The Cayley's composition table under binary operation modulo 4 of multiplication is,

$\times_4$	0	1	2	3
0	0	0	0	0
1	0	<u>1</u>	2	3
2	0	2	0	2
3	0	3	2	<u>1</u>

from above table,

1) Closure:-

for all $a, b \in Z_4$
 $a \times_4 b \in Z_4$

$\therefore Z_4$ is closed under $\times_4$

2) Associative :-

for $a, b, c \in \mathbb{Z}_4$

$$a \times_4 (b \times_4 c) = (a \times_4 b) \times_4 c$$

$\therefore \mathbb{Z}_4$ is associative.

3) Identity :-

Here, 1 is identity.

$$\text{i.e., } 0 \times_4 1 = 0$$

$$1 \times_4 1 = 1$$

$$2 \times_4 1 = 2$$

$$3 \times_4 1 = 3$$

4) Existence of inverse :-

* Abelian Group :-

A group $(G, *)$ is said to be abelian group if it satisfies commutative property.

i.e.,

for any $a, b \in G$
 $a * b = b * a$

e.g. $(\mathbb{Z}, +), (\mathbb{R}, +), (\mathbb{Z}, \times), (\mathbb{Z}_m, t_m)$

⇒ Flow-chart :-

Algebraic structure $\longrightarrow$ closure



Semi-group $\longrightarrow$ Closure + Associative



Monoid $\longrightarrow$ Closure + Associative + identity



Group $\longrightarrow$ Monoid + Inverse



Abelian Group $\longrightarrow$ Group + Commutative

Q. 1. Prove that $\mathbb{Z}_4 = \{0, 1, 2, 3\}$ is an Abelian group w.r.t t_4

→ Here,

$$\mathbb{Z}_4 = \{0, 1, 2, 3\}$$

(The caylay's composition table under binary operation modulo 4 of addition is,

$+_4$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

from above table,

1) Closure:-

— for all $a, b \in \mathbb{Z}_4$

$$a +_4 b \in \mathbb{Z}_4$$

$\therefore \mathbb{Z}_4$ is closed under $+_4$

2) Associative:-

— for $a, b, c \in \mathbb{Z}_4$

$$a +_4 (b +_4 c) = (a +_4 b) +_4 c$$

$\therefore \mathbb{Z}_4$ is an associative.

3) Identity:-

Here, 0 is identity.

$$\text{i.e., } 0 +_4 0 = 0$$

$$1 +_4 0 = 1$$

$$2 +_4 0 = 2$$

$$3 +_4 0 = 3$$

4) Existence of inverse:-

$$0 +_4 0 = 0$$

$$1 +_4 3 = 0$$

$$2 +_4 2 = 0$$

$$3 +_4 1 = 0$$

$\therefore$ Inverse exist for every element of $\mathbb{Z}_4$.

5) Commutative:-

$$0 +_4 0 = 0$$

$$1 +_4 2 = 2 +_4 1 = 3$$

$$2 +_4 3 = 3 +_4 2 = 1$$

$$3 +_4 1 = 1 +_4 3 = 0$$

Thus $(\mathbb{Z}_4, +_4)$ is an Abelian Group.

Q.2. Prove that $\mathbb{Z}_{10} = \{1, 3, 7, 9\}$ is an Abelian group w.r.t $\times_{10}$

H.W.

Q. 3. Show that $G = \{0, 1, 2, 3, 4, 5\}$ under $+$ is an Abelian group under $+$.

H.W.

Q. 4. Show that the cube roots of unity is an Abelian group under multiplication.

Here, $G = \{1, \omega, \omega^2\}$

→
Cube root of unity $= 1, \frac{-1 \pm i\sqrt{3}}{2}$
 $= 1, \omega, \omega^2$

The Cayley's composition table can be given as,

X	1	ω	ω^2
1	$\boxed{1}$	ω	ω^2
ω	ω	ω^2	$\boxed{\omega^3=1}$
ω^2	ω^2	$\boxed{\omega^2=1}$	ω

Now,

i) Closure :-

for any $a, b \in G$
 $a * b \in G$

ii) Associativity :-

for any $a, b, c \in G$
 $a * (b * c) = (a * b) * c$
 $1 * (\omega * \omega^2) = (1 * \omega) * \omega^2 = 1$

$\therefore G$ follows associativity.

iii) Existence of Identity:-

Under multiplication the identity is 1,
 $1 \in G$.

Thus, identity exist

iv) Existence of Inverse:-

$$(1)^{-1} = \omega^3 = 1$$

$$(\omega)^{-1} = \omega^2$$

$$(\omega^2)^{-1} = \omega$$

Thus,

for every element inverse exist.

v)

$$1 \times \omega = \omega \times 1 = \omega$$

$$\omega \times \omega^2 = \omega^2 \times \omega = \omega^3 = 1$$

$$1 \times \omega^2 = \omega^2 \times 1 = \omega^2$$

Thus,

the cube roots of unity form an abelian group.

Ex.

Show that fourth roots of unity form an abelian group under multiplication.

* Cyclic Group :-

A group $(G, *)$ is called cyclic if there exist an element $a \in G$ such that every element of G can be generated by a (i.e., written as a^n) for some integer n .

$$\therefore G = \{a^n \mid n \in \mathbb{Z}\}$$

We say G is generated by a @ a is generator of G .

Note :- $\Rightarrow$ If binary operation is addition then
 $a^n = a + a + a + a + \dots n \text{ times} = na$.

$\Rightarrow$ If binary operation is multiplication then
 $a^n = a \times a \times a \times a \times \dots n \text{ times}$

Q.1. Prove that $(G, \times)$ is cyclic group where
 $G = \{1, \omega, \omega^2\}$

$\rightarrow (G, \times)$ is a group where $G = \{1, \omega, \omega^2\}$
 We know that it is an abelian group.

To prove it is cyclic define generators

The caylay's composition table is given as,

$\times$	1	ω	ω^2
1	1	ω	ω^2
ω	ω	ω^2	1
ω^2	ω^2	1	ω

i) for 1 :-

$$1 \times 1 = 1$$

$$1^2 = 1 \times 1 = 1$$

$$1^3 = 1 \times 1 \times 1 = 1$$

$\therefore 1$ is not a generator.

ii) for ω :-

$$\omega = \omega \times 1 = \omega$$

$$(\omega)^2 = \omega \times \omega = \omega^2$$

$$(\omega^3) = \omega \times \omega^2 = 1$$

$\therefore \omega$ is generator with order 3.

iii) for ω^2 :-

$$(\omega^2)^1 = \omega^2$$

$$(\omega^2)^2 = \omega^2 \times \omega^2 = \omega$$

$$(\omega^2)^3 = \omega^2 \times \omega^2 \times \omega^2 = (\omega^2)(\omega^2)^2 = 1$$

$\therefore \omega^2$ is ^{also} generator with order 3.

Thus,

$$G = \langle \omega, \omega^2 \rangle$$

$\therefore \omega$ & ω^2 are generators of G .

$\therefore (G, \times)$ is cyclic group

Q.2. Prove that $G = \{1, -1, i, -i\}$ is cyclic group under multiplication and find all generators.

Given,

$$G = \{1, -1, i, -i\}$$

The Cayley's composition table can be given as,

$\times$	1	-1	i	-i
1	1	-1	i	-i
-1	-1	1	-i	i
i	i	-i	-1	1
-i	-i	i	1	-1

G is closed under multiplication also, associativity holds for any $a, b, c \in G$.

1 is multiplicative identity and 1 belongs to G , Therefore, identity exists.

Also inverse for every element exists.

$\therefore G$ is group.

Now,

To find generators

1 and (-1) are not generators.

(1) For i :-

$$(i)^1 = i$$

$$(i)^2 = -1$$

$$(i)^3 = i \times i^2 = i \times (-1) = -i$$

$$(i)^4 = i \times i = (-1) \times (-1) = 1$$

$\therefore i$ is generator with order 4.

② Also for i^2 :-

~~$$\left(\frac{0}{1}\right)' = \frac{0^2}{1^2}$$~~

~~$$\left(\frac{0}{1}\right)^2 = \frac{0^2}{1^2} \times \frac{0^2}{1^2} = (-1) \times (-1) = 1$$~~

~~$$\left(\frac{0}{1}\right)^3 = \frac{0^2}{1^2} \times \left(\frac{0}{1}\right)^2 = (-1) \times (-1) =$$~~

$$(-i)' = -i$$

$$(-i)^2 = (-i) \times (-i) = i^2 = -1$$

$$(-i)^3 = (-i)^2 \times (-i) = -1 \times -i = i$$

$$(-i)^4 = (-i)^2 \times (-i)^2 = -1 \times -1 = 1$$

$\therefore -i$ is generator with order 4

Thus, i & $-i$ are generator of G .

$\therefore G$ is cyclic group

Q.3. Show that $G = \{0, 1, 2, 3, 4, 5\}$ under $+$ is cyclic and find its generator.

→ Given,

$$G = \{0, 1, 2, 3, 4, 5\}$$

We know that,

$(\mathbb{Z}_m, +_m)$ is always a group.

under addition '0' is identity.

The caylay's composition table

$+$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

① for 1 :-

~~$1(0) = 0$~~

$1(1) = 1$

$1(2) = 1 + 1 = 2$

$1(3) = 3$

$1(4) = 4$

$1(5) = 5$

Therefore, 1 is generator with order = 6.

② for 2 :-

$2(1) = 2$

$2(2) = 4$

$2(3) = 6 = 0$

$2(4) = 8 = 2$

$2(5) = 10 = 4$

$2(6) = 12 = 0$

∴ 2 is not a generator.

③ for 3 :-

$3(1) = 3$

$3(2) = 6 = 0$

$3(3) = 9 = 3$

$3(4) = 12 = 0$

$3(5) = 15 = 3$

∴ 3 is not a generator.

for 4 :-

$4(1) = 4$

$4(2) = 8 = 2$

$4(3) = 12 = 0$

$4(4) = 16 = 4$

$4(5) = 20 = 2$

∴ 4 is not a generator.

for 5

$$5(1) = 5$$

$$5(2) = 10 = 4$$

$$5(3) = 15 = 3$$

$$5(4) = 20 = 2$$

$$5(5) = 25 = 1$$

$$5(6) = 30 = 0$$

$\therefore 5$ is generator with order 6.

Thus, 1 & 5 are generator

$\therefore (G_6, +_6)$ is cyclic group

Q.4. Prove that $\{0, 1, 2, 3\}$ is cyclic group under $+_4$

(or)

Prove that $(Z_4, +_4)$ is cyclic group.

(or)

Show that Z_4 is cyclic group under $+_4$ and thus find all its generators.

* Properties of Cyclic Group :-

- 1) Every cyclic group is abelian.
- 2) Order of generators is same as order of cyclic group.
- 3) If a is generator then a^{-1} is also generator.
- 4) If a is generator then a^p is also generator if $(p, n) = 1$ i.e., p is relatively prime to n .

Q. find all generators of cyclic group $G = \{1, 2, 3, 4\}$ under $\times_5$

* Permutation Group :-

Let P be the finite set have n distinct elements. is a one-one and onto from P to itself. is called permutation of degree n .

eg:
$$P = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

$$= (1\ 2\ 3)$$

The collection of all permutations of a non-empty set A forms a group under operation composition it is called permutation group denoted by S_n

* Product of Permutations :-

The product of two permutations f & g of same degree denoted by fg can be given as
 If

$$f = \begin{pmatrix} a_1 & a_2 & a_3 & a_4 \\ b_1 & b_2 & b_3 & b_4 \end{pmatrix}$$

$$g = \begin{pmatrix} b_1 & b_2 & b_3 & b_4 \\ c_1 & c_2 & c_3 & c_4 \end{pmatrix}$$

$$fg = \begin{pmatrix} a_1 & a_2 & a_3 & a_4 \\ c_1 & c_2 & c_3 & c_4 \end{pmatrix}$$

* Transposition :-

Any cycle of length 2 is called transposition

eg (1 2), (2 3)

Q. 1 find $f \circ g$ and show that is not commutative for its

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad g = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{pmatrix}$$

→ Given,

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \& \quad g = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{pmatrix}$$

$$f \circ g = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{pmatrix}$$

$$\therefore f \circ g = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \text{--- (1)}$$

&

$$g \circ f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \text{--- (2)}$$

from (1) & (2),

$$f \circ g \neq g \circ f$$

$\therefore$ It is not commutative.

* Even and Odd permutation :-

⇒ If a cycle is decomposed in even number of transpositions then it is called even permutation.

⇒ If a cycle is decomposed in odd number of transposition then it is called odd permutation.

Note :- Decomposition of cycle $(1\ 2\ 3) = (1\ 2)(1\ 3)$

Q. 2. Examine odd/Even for $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 2 & 5 & 4 & 3 & 6 & 1 & 7 & 9 & 8 \end{pmatrix}$

→ Given permutation can be written in the form of cycle as,

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 2 & 5 & 4 & 3 & 6 & 1 & 7 & 9 & 8 \end{pmatrix}$$

$$f = (1\ 2\ 5\ 6)(3\ 4)(7)(8\ 9)$$

$$f = (1\ 2)(1\ 5)(1\ 6)(3\ 4)(8\ 9)$$

Don't consider single length cycle

→ decomposed here

Here we have 5 transpositions
 $\therefore f$ is odd permutation.

H.W.
Q. 3. IF $f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 5 & 1 \end{pmatrix}$ & $g = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 4 & 5 & 3 \end{pmatrix}$ then find $f \circ g$, $g \circ f$ & write their cycles also determine whether odd or even.

H.W.
Q. 4. $(1\ 2\ 3) \cdot (5\ 6\ 4\ 1)$ find cycles of $f \circ g$ & $g \circ f$.

H.W.
Q. 5. $f = (1\ 6\ 5\ 2\ 7\ 3\ 4\ 8)$
 $g = (1\ 5\ 2\ 4\ 6\ 8) \cdot (3\ 7)$ find cycles of $f \circ g$ & $g \circ f$

H.W.
Q. 6. $(1\ 3\ 2\ 5\ 6) \cdot (2\ 3) \cdot (4\ 6\ 5\ 1\ 2)$
 if all are cycles of length 6.

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$(1\ 3\ 2\ 5\ 6) \cdot (2\ 3) \cdot (4\ 6\ 5\ 1\ 2)$ if all are cycles of length 6.

Let $f = (1\ 3\ 2\ 5\ 6) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 5 & 2 & 4 & 6 & 1 \end{pmatrix}$

$g = (2\ 3) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 3 & 2 & 4 & 5 & 6 \end{pmatrix}$

$h = (4\ 6\ 5\ 1\ 2) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 3 & 6 & 1 & 5 \end{pmatrix}$

$\therefore fogoh = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 1 & 3 & 6 & 5 & 2 \end{pmatrix} = (1\ 4\ 6\ 2) (3) (5)$

* Homomorphism & Isomorphism of semi-group :-

1) Homomorphism of semi-group :-

Let, $(S, *)$ & $(T, *)$ the two semi-groups.

A $f: S \rightarrow T$ is called homomorphism from $(S, *)$ to $(T, *)$ if,
 $\rightarrow f(a * b) = f(a) * f(b) \quad \forall a, b \in S.$

2) Isomorphism of semi-group :-

Let, $(S, *)$ & $(T, *')$ the two semi-groups.

A $f: S \rightarrow T$ is called isomorphism from $(S, *)$ to $(T, *')$ if,

① $\rightarrow f(a * b) = f(a) *' f(b) \quad \forall a, b \in S.$

② f is one-one & onto (bijective)

Note:- If f is homomorphic & bijective then S is isomorphic.

* Coset :-

Let H be a sub-group of a group G and $a \in G$ then the set

1) $aH = \{ah \mid h \in H, a \in G\}$ is called left coset of h in G .

2) $Ha = \{ha \mid h \in H, a \in G\}$ is called right coset of h in G .

Thus corresponding to every element of G we have left and right cosets of H in G .

Also,

$$eH = He = H.$$

example: Let $G = (\mathbb{Z}, +)$ & $H = 2\mathbb{Z} = \{0, \pm 2, \pm 4, \dots\}$ be the subgroup of G then find all the cosets of H in G .

Given,

$$G = \mathbb{Z}^+$$

$$H = 2\mathbb{Z} = \{0, \pm 2, \pm 4, \pm 6, \dots\}$$

To find the cosets of H in G ,

$$\therefore H + 0 = \{0, \pm 2, \pm 4, \dots\}$$

$$H + 1 = \{\pm 1, \pm 3, \pm 5, \dots\}$$

$$H + 2 = \{0, \pm 2, \pm 4, \pm 6, \dots\}$$

$$H + 3 = \{\pm 1, \pm 3, \pm 5, \dots\}$$

$$H + 4 = \{0, \pm 2, \pm 4, \dots\}$$

$$H + 5 = \{\pm 1, \pm 3, \pm 5, \dots\}$$

As the operation is addition $H + a = a + H$

$\therefore H$ & $H + 1$ are two distinct cosets of G .

Q. 2. Find all cosets of $3\mathbb{Z}$ in $(\mathbb{Z}, +)$

Given,

$$G = (\mathbb{Z}, +)$$

$$H = 3\mathbb{Z} = \{0, \pm 3, \pm 6, \pm 9, \dots\}$$

To find cosets of H ,

$$\therefore H+0 = \{0, \pm 3, \pm 6, \pm 9, \dots\}$$

$$H+1 = \{\dots, -8, -5, -2, 1, 4, 7, 10, \dots\}$$

$$H+2 = \{\dots, -7, -4, -1, 2, 5, 8, 11, \dots\}$$

$$H+3 = \{0, \pm 3, \pm 6, \pm 9, \dots\}$$

$$H+4 = \{\dots, -8, -5, -2, 1, 4, 7, 10, \dots\}$$

$$H+5 = \{\dots, -7, -4, -1, 2, 5, 8, 11, \dots\}$$

As the operation of addition is

$$H+1 = H+4$$

$$H+2 = H+5$$

$$H+0 = H+3$$

Thus, $H+0 = H+3 = H+6 = \dots$

$$H+1 = H+4 = H+7 = \dots$$

$$H+2 = H+5 = H+8 = \dots$$

$\therefore$ We have three distinct cosets $H, H+1, H+2$ in G .

Note:- For $n\mathbb{Z}$ in $(\mathbb{Z}, +)$ we have 'n' distinct cosets.

Q. 3. Find all cosets of $H = \{0, 4\}$ in $G = (\mathbb{Z}_8, +_8)$

* Normal Subgroup :-

A sub group H of a group G is called Normal subgroup if,

$$ah = ha \quad \forall a \in G, h \in H$$

example :-

Q.1. $G = \{1, -1, i, -i\}$ with $(G, \times)$ & $H = \{1, -1\}$. Then prove that, H is normal subgroup of G . ~~solution~~

Given,

$$G = \{1, -1, i, -i\}$$

&

$$H = \{1, -1\}$$

To prove normal subgroup we find left and right cosets.

Right coset.

$$H \cdot 1 = \{1, -1\}$$

$$H \cdot -1 = \{-1, 1\}$$

$$H \cdot i = \{i, -i\}$$

$$H \cdot -i = \{-i, i\}$$

Left coset.

$$1 \cdot H = \{1, -1\}$$

$$-1 \cdot H = \{-1, 1\}$$

$$i \cdot H = \{i, -i\}$$

$$-i \cdot H = \{-i, i\}$$

$$\therefore H \cdot 1 = 1 \cdot H ; H \cdot -1 = -1 \cdot H ; i \cdot H = H \cdot i ; -i \cdot H = H \cdot -i ;$$

Thus,

$$ah = ha \quad \forall a \in G, h \in H.$$

$$\Rightarrow \boxed{aH = Ha}$$

$\therefore H$ is normal subgroup of G .

$$\Rightarrow \boxed{H \trianglelefteq G}$$

Theorem :-

A subgroup H of group G is normal if and only if $g^{-1}hg \in H \rightarrow \forall h \in H, g \in G$.

Proof :-

Let H be normal sub-group of G .

Let $h \in H, g \in G$

$$\Rightarrow Hg = gH$$

Now,

$$gh = hg \in Hg$$

$$\therefore hg = gh, \text{ for } h, \in H.$$

multiply b.s. by g^{-1} .

$$g^{-1}hg = g^{-1}gh, = eh, = h,$$

$$\therefore g^{-1}hg = h, \in H$$

Let, H be such that.

$$g^{-1}hg \in H$$

for any $a \in G, h \in H$

$$a^{-1}ha \in H$$

premultiply by a on both side.

$$a \cdot a^{-1}ha \in aH.$$

$$eha \in aH.$$

$$ha \in aH.$$

$$\Rightarrow \text{But } ha \in Ha$$

$$\Rightarrow Ha \subseteq aH$$

$$\therefore Ha = aH$$

Thus, H is normal sub-group of G .

Rings.~~Thus, H is normal~~* Rings :-

An algebraic structure $(R, +, \cdot)$ is a ring if R is a non-empty set and '+' & ' $\cdot$ ' are binary operations such that,

- 1) $(R, +)$ is Abelian group
- 2) $(R, \cdot)$ is semi-group.
- 3) It should satisfy left & right distributive law.

i.e.,

$$a \cdot (b + c) = ab + ac$$

$$(a + b) \cdot c = ac + bc$$

→ Left distributive law

→ right ——— //

* Types of Rings :-1) Commutative ring :-

Let an algebraic structure $(R, +, \cdot)$ be a ring, then R is called commutative ring if its multiplication is commutative.

i.e., $a \cdot b = b \cdot a$ for $\forall a, b \in R$.

⇒ Ring with identity / unity :-

Ring $(R, +, \cdot)$ is called ring with unity element for multiplication if there exist $1 \in R$ such that

$$\forall a \in R \quad \boxed{1 \cdot a = a \cdot 1 = a}$$

3) Commutative ring with unity :-

If a ring have multiplication identity and it is commutative then ring is called commutative ring with unity.

4) Zero ring / Null ring / Trivial ring :-

The set are consisting of single element zero with operation '+' & '.' defined as $0+0=0$ & $0 \cdot 0=0$. then R is called

5) Fields :-

A ring $(R, +, \cdot)$ with atleast two element is called field if it is commutative ring with unity. and its each non zero element possessess multiplicative inverse.

It is denoted by $(F, +, \cdot)$